specific data points that tell you if your application is providing value to your consumers. Because an entire chapter is devoted to high-level metrics to track, I'll jump right into the elements that affect consumer satisfaction and business success criteria and where to get them.

8.4.1. Reliability and availability

Reliability and availability are often grouped together when the code "ilities" are discussed. I used to think they were the same thing, but when I measure them they definitely point to different but closely related factors.

Availability is the measure of how much time your application is functioning relative to how frequently your consumers want to use it. If you have a web application for a global user base, then your availability needs to be as close to 100% as possible because people from different time zones will be logging in and using it as the globe spins. In addition, you may have contracts with partners that specify service level agreements (SLAs) that bind you to supporting availability times of 99.999%, or about 4 hours a month. Conversely, if your application is used in a retail location that is open from 7 a.m. to 9 p.m. seven days a week, it probably doesn't matter if it's not available from 9:01 p.m. to 6:59 a.m. every day. Availability can be measured by the following metrics:

- ***Uptime*** —What percentage of the time your application is functioning.
- ***Page response time*** —If your application is so slow that pages aren't loading in the time your consumer expects, you can consider it unavailable.

Reliability means how consistently your application does what it's supposed to do. If you have intermittent issues with your application, it's not reliable. For example, if you have an e-commerce application where you can't add items to the shopping cart when the site is under heavy load, it's not very reliable. You can measure reliability with the following metrics:

- ***Mean time between failures*** —How frequently your application breaks for your consumers.